MIL-S-11578D
23 March 1978
SUPERSEDING
MIL-S-11578C
6 August 1970

MILITARY SPECIFICATION

# SHAKING MACHINE, TESTING SIEVE, LABORATORY, PORTABLE, HAND-OPERATED, CAPACITY 6 FULL-HEIGHT OR 13 HALF-HEIGHT, 8-INCH STANDARD SIEVES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a hand-operated, portable shaking machine including cover and pan.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| QQ-S-781 | - Strapping, Steel, and Seals. |
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood. |
| PPP-B-636 | - Boxes, Shipping, Fiberboard. |

FSC 6635

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

| | |
|---|---|
| MIL-P-116 | - Preservation-Packaging, Methods of. |
| MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-130 | - Identification Marking of US Military Property. |
| MIL-STD-1188 | - Commercial Packaging of Supplies and Equipment. |

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 11 - Wire-Cloth Sieves for Testing Purposes.
E 323 - Perforated-Plate Sieves for Testing Purposes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)



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3. REQUIREMENTS

3.1 Description. The shaking machine shall consist essentially of a sieve carriage and support, an oscillating mechanism, and a base similar to that shown on Figures 1 and 2 (see 6.3).

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3 Safety.

3.3.1 Moving parts. All moving parts of the oscillating mechanism, except the flywheel, shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard.

3.3.2 Lubrication points. All points requiring lubrication during operation shall have fittings located so as to be accessible without hazardous exposure to personnel performing these operations.

3.4 Sieve carriage and support. The sieve carriage and support shall be aluminum or magnesium alloy, or when specified (see 6.2), iron or steel. The sieve carriage and support shall be removable for disassembly by one man using only common hand tools. The carriage shall hold from one to not less than six full-height or from one to not less than 13 half-height 8-inch sieves, including sieve cover and pan, conforming to ASTM E 11 and ASTM E 323. The carriage shall be suspended from the support and shall engage the oscillating mechanism.

3.5 Oscillating mechanism. The oscillating mechanism shall operate through a hand-cranked gear or belt-driven flywheel. The sieve platform shall move in an eccentric orbit and shall receive not less than two or not more than eight impacts during each orbit. The flywheel shall be approximately 10 inches in diameter. The flywheel shall weigh not less than 6 pounds or not more than 10 pounds. The oscillator shaft shall be provided with bearings. The sieve carriage shall make one complete oscillation (carriage bed traveling a minimum distance of 2-3/4 inches and returning to its original position) for each revolution of the flywheel. The handcrank shall drive the flywheel through a ratio of not more than 4 to 1, or not less than 3 to 1. The crank shall be removable for storage.



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3.6 Base. The base shall be cast iron or the same material as the carriage and shall provide rigid support for the oscillating mechanism and the sieve carriage support. Provisions shall be made for bolting the base to a worktable.

3.7 Performance. When tested as specified in 4.4.2, the shaking machine shall perform in such a manner that not more than 1 gram change will occur in the contents of the sieves or sieve pan when weighed.

3.8 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.9 Identification marking. The shaking machine shall be identified in accordance with MIL-STD-130. The marking shall be applied to the shaking machine on plates conforming to MIL-P-514, Type III, Composition D.

3.10 Treatment and painting. Treatment and painting shall be at the contractor's option.

3.11 Workmanship. All parts, components, and assemblies of the shaking machine including castings, forgings, molded parts, stampings, and machined surfaces shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.



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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3.2 Examination.

4.3.2.1 Samples. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective.

4.3.3 Test.

4.3.3.1 Samples. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The shaking machine shall be examined as specified herein for the following defects:

101. Material not as specified.
102. Safety provisions not as specified.
103. Components missing or not as specified.
104. Lubrication not as specified.
105. Identification marking missing, incorrect, or illegible.
106. Workmanship not as specified.

4.4.2 Performance test. Five half-height sieves conforming to ASTM E 11, sieve designation numbers 10, 20, 40, 100, and 200, with the top cover and sieve pan, shall be put into the sieve carriage and the shaking machine charged with approximately 400 grams of sand. The gradation of the sand shall be such that approximately the same weight is retained on each sieve after the test is completed. The handcrank shall be operated at not less than 70 revolutions per minute (rpm) for 15 minutes. The sieves shall then be removed and the material retained on each sieve and in the sieve pan weighed and the weight recorded.



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The sand from each sieve and pan shall then be returned to its original charging position and the sieving operation shall be repeated at the same speed for 1 minute. The sieves shall be removed, and the weight of sand on each sieve and pan determined and recorded. The change in weight of the sand on each sieve and the sieve pan shall be determined. Nonconformance to 3.7 shall constitute failure of this test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

| | | |
|---|---|---|
| 107. | Materials, methods, or containers not as specified. Each incorrect material, method, or container shall be considered one defect. |
| 108. | Unprotected surface not coated with preservative as specified. |
| 109. | Box closure not Method V as specified. |
| 110. | Strapping not as specified. |
| 111. | Marking illegible, incomplete, incorrect, or missing. |

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial as specified (see 6.2).

5.1.1 Level A. Each shaking machine shall be disassembled as necessary to provide a pack of minimum size. Unprotected metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116. Each shaking machine shall then be preserved in accordance with MIL-P-116, Method III using a close-fitting box conforming to PPP-B-636, W5c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.



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5.1.2 Level B. Each shaking machine shall be preserved as specified in 5.1.1 for Level A, except that the box shall be Type CF, Class Domestic, Variety SW, Grade 200, and Method V closure will not be required.

5.1.3 Commercial. Each shaking machine shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. Shaking machines preserved as specified in 5.1 shall be packed together in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Shaking machines shall be packed as specified in 5.2.1 for Level A, except that boxes shall be Domestic Type and strapping shall be Finish A.

5.2.3 Commercial. Shaking machines preserved as specified in 5.1 shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The shaking machine is intended to be used for the determination of grain-size distribution of a soil.



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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When an iron or steel sieve carriage and support is required (see 3.4).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(d) When Finish A strapping is required (see 5.2.1).

6.3 Information figures. Figures 1 and 2 show two types of shaking machines which have been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of other shaking machines which conform to this specification.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

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Army - ME
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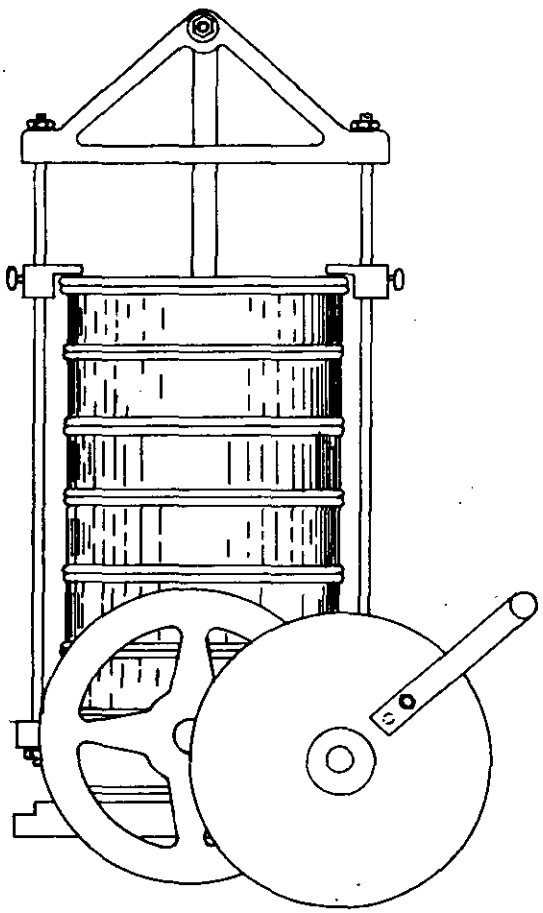


FIGURE 1. Shaking machine, testing sieve.

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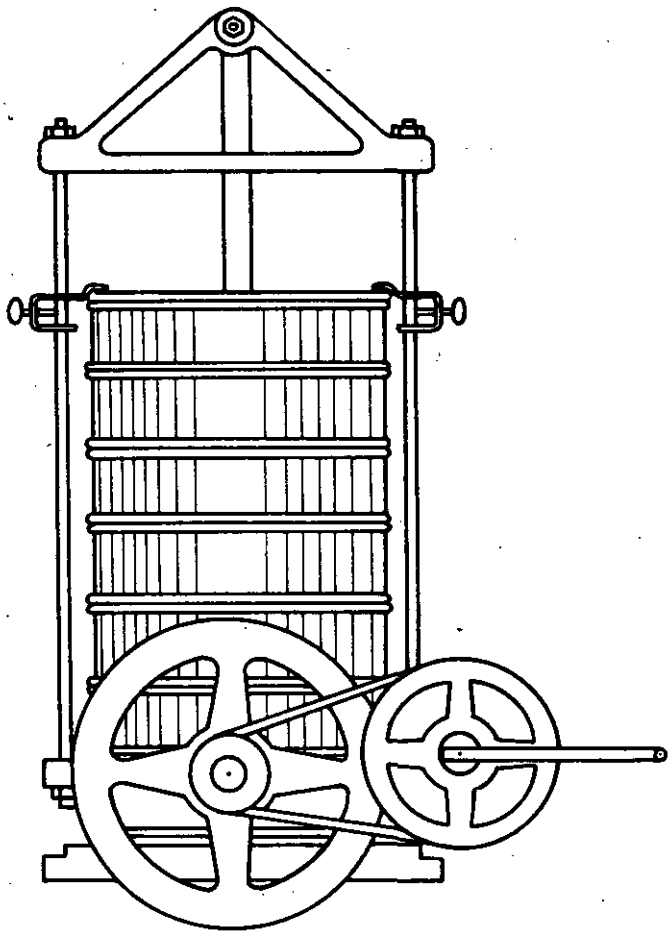


FIGURE 2. Shaking machine, testing sieve.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-S-11578D Shaking Machine, Testing Sieve, Laboratory, Portable, Hnd-Operated, Capacity 6 Full-Height or 13 Half-Height, 8-In Standard Sieves

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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